

# **Iowa State University**

## **Laboratory Report #4**

Course: AERE 421: Undergraduate Advanced Flight Structures

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Section 1 Group 15  
Group Members: Corbin Marcello, Lucas Tavares Vasconcellos  
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## **PART 1**

### **Introduction**

For part 1, we are prompted to create and analyze an aircraft fuselage with a circular window, modeled as a half cylinder with a circular hole in the middles. The fuselage material is a layered composite with  $t = 125\text{e-}6\text{m}$  and ply sequence  $[0_4/45_2/-45_2]_s$ . The elastic constant and failure characteristics are defined in the following tables:

| <b>Elastic Constants</b>      |                               |                               |
|-------------------------------|-------------------------------|-------------------------------|
| $E_x = 154 \text{ GPa}$       | $E_y = 14.79 \text{ GPa}$     | $E_z = 14.79 \text{ GPa}$     |
| $\nu_{yz} = 0.3$              | $\nu_{zx} = 0.175$            | $\nu_{xy} = 0.175$            |
| $\mu_{yz} = 3.37 \text{ GPa}$ | $\mu_{zx} = 7.48 \text{ GPa}$ | $\mu_{xy} = 7.48 \text{ GPa}$ |

| <b>Failure Loads</b>         |                             |                             |
|------------------------------|-----------------------------|-----------------------------|
| $F_{1t} = 1500 \text{ MPa}$  | $F_{2t} = 40 \text{ MPa}$   | $F_{3t} = 40 \text{ MPa}$   |
| $F_{1c} = -1500 \text{ MPa}$ | $F_{2c} = -246 \text{ MPa}$ | $F_{3c} = -246 \text{ MPa}$ |
| $F_{xy} = 68 \text{ MPa}$    | $F_{xz} = 68 \text{ MPa}$   | $F_{yz} = 68 \text{ MPa}$   |

### **Objectives**

- Displacement plot.
- Output contour plots of:
  - Graphical representation of the total lay-up.
  - Stresses and strains.
  - Overall failure index (Inverse of Tsai-Wu strength ratio index).
  - Failure index for each ply orientation.

## Results

### Composite Fuselage with window (Part 1)

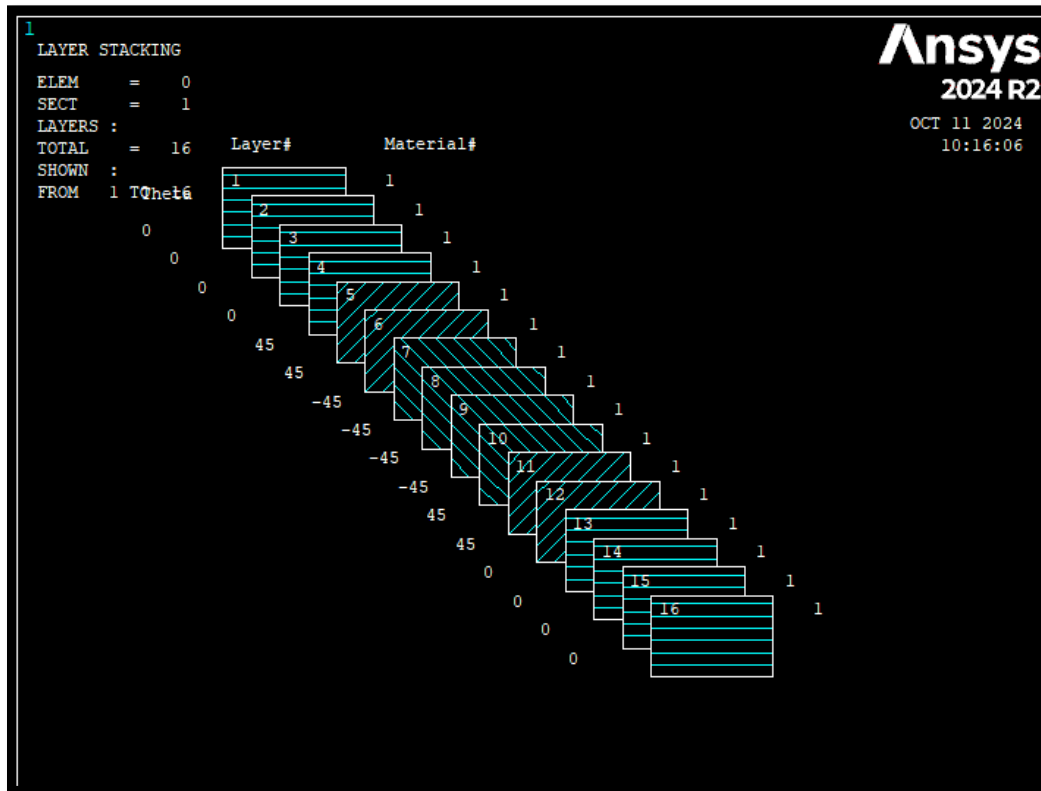


Figure 1: Composite Layup

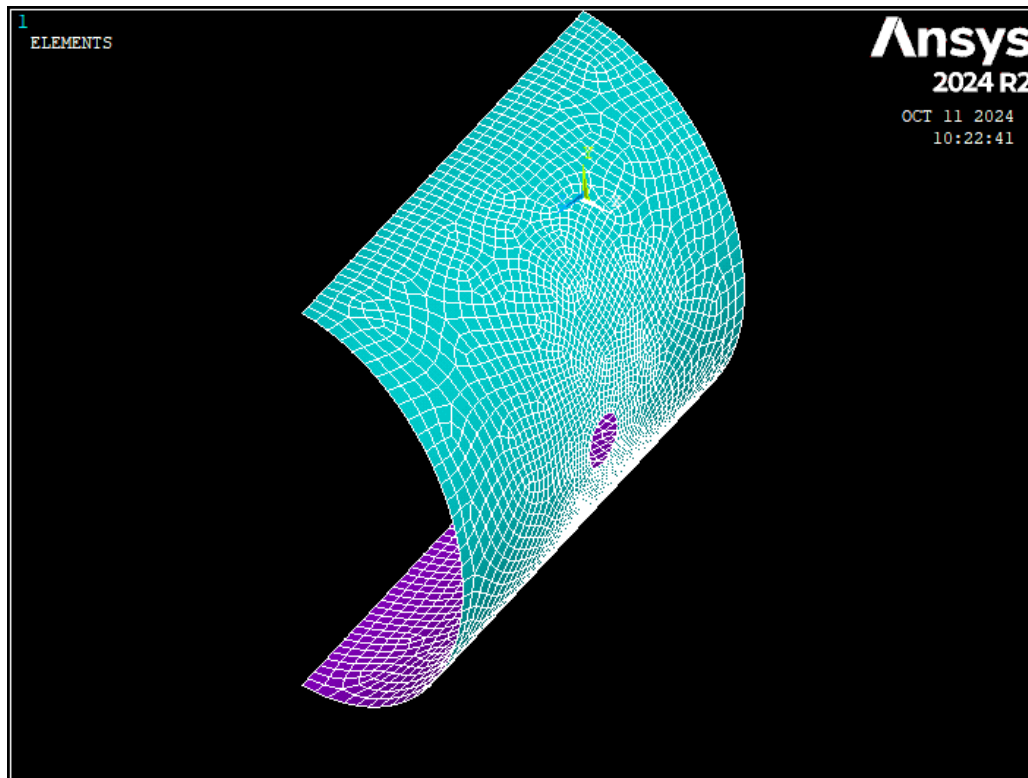


Figure 2: General Geometry of the Fuselage (Isometric View)

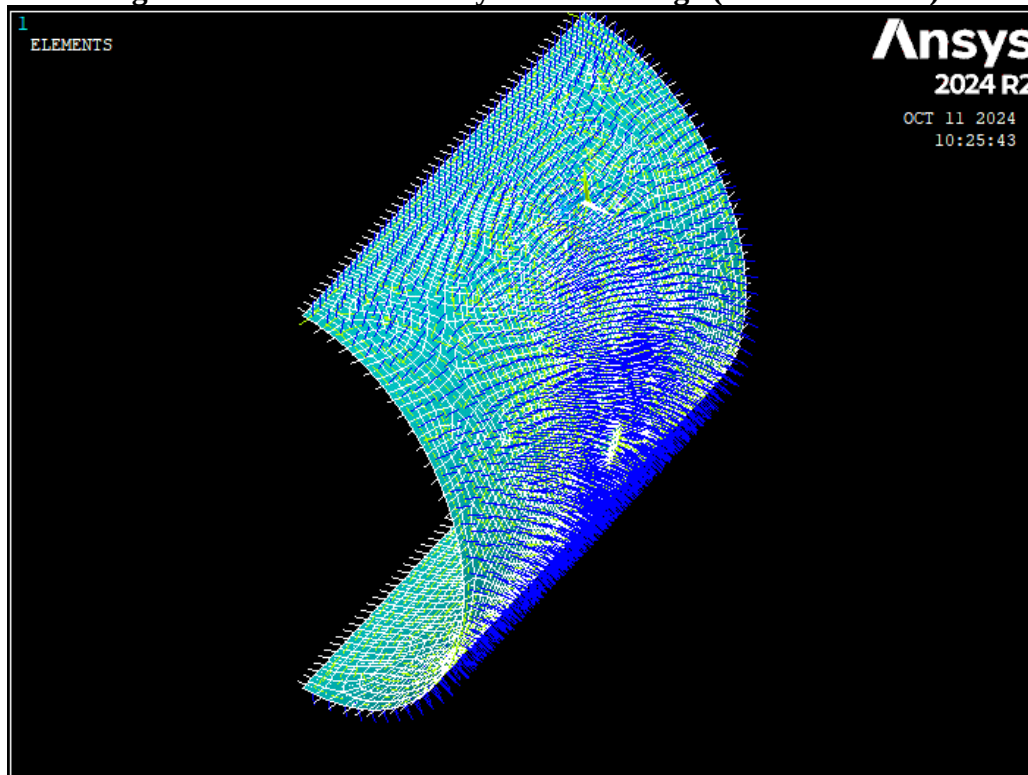
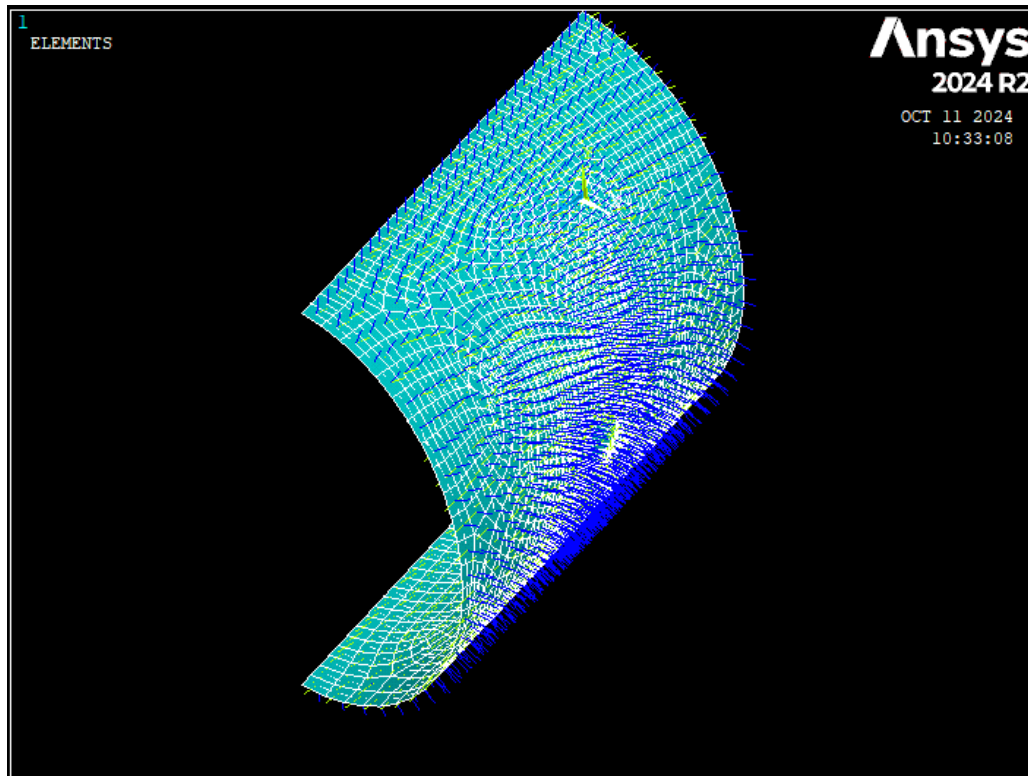
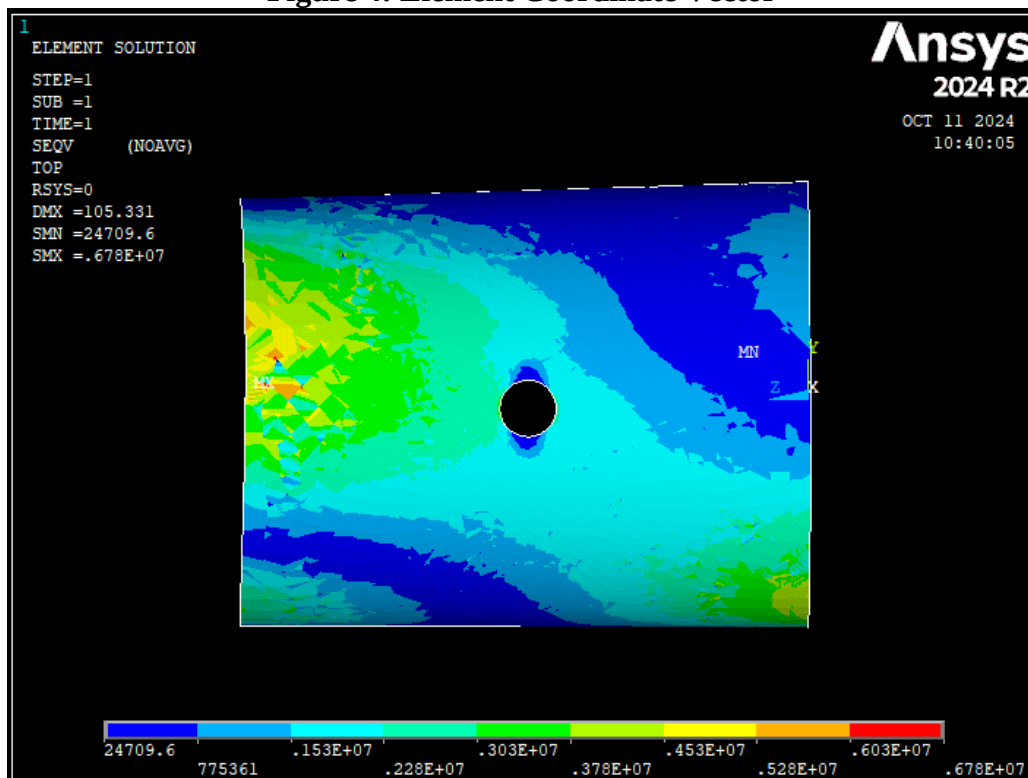
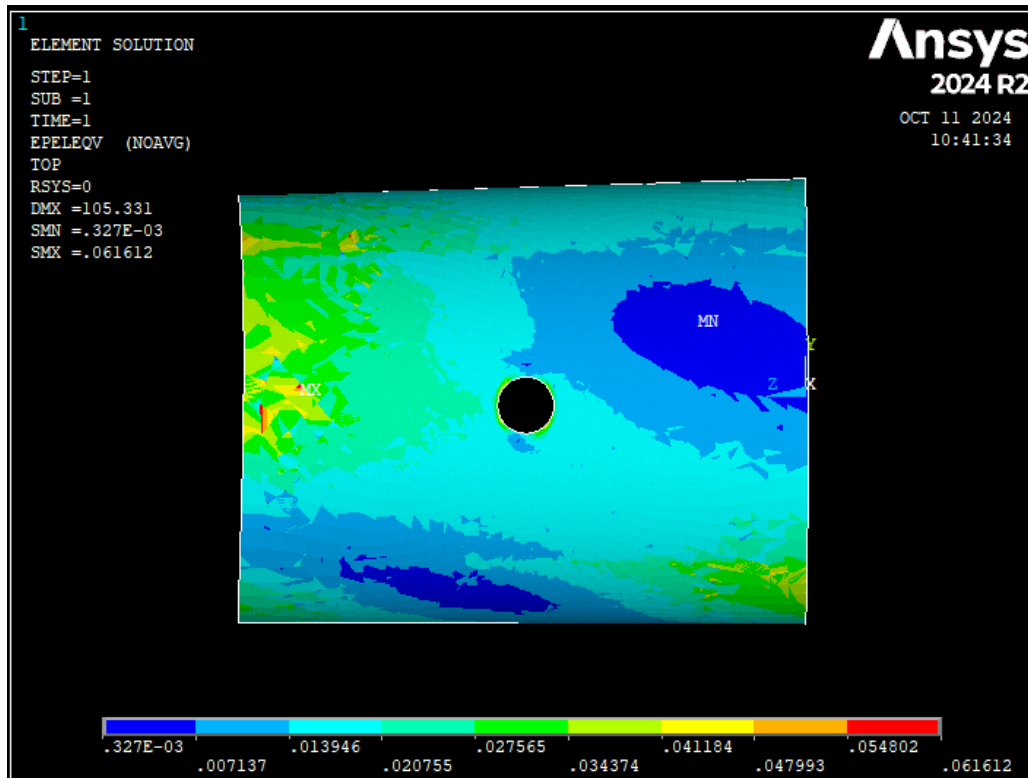
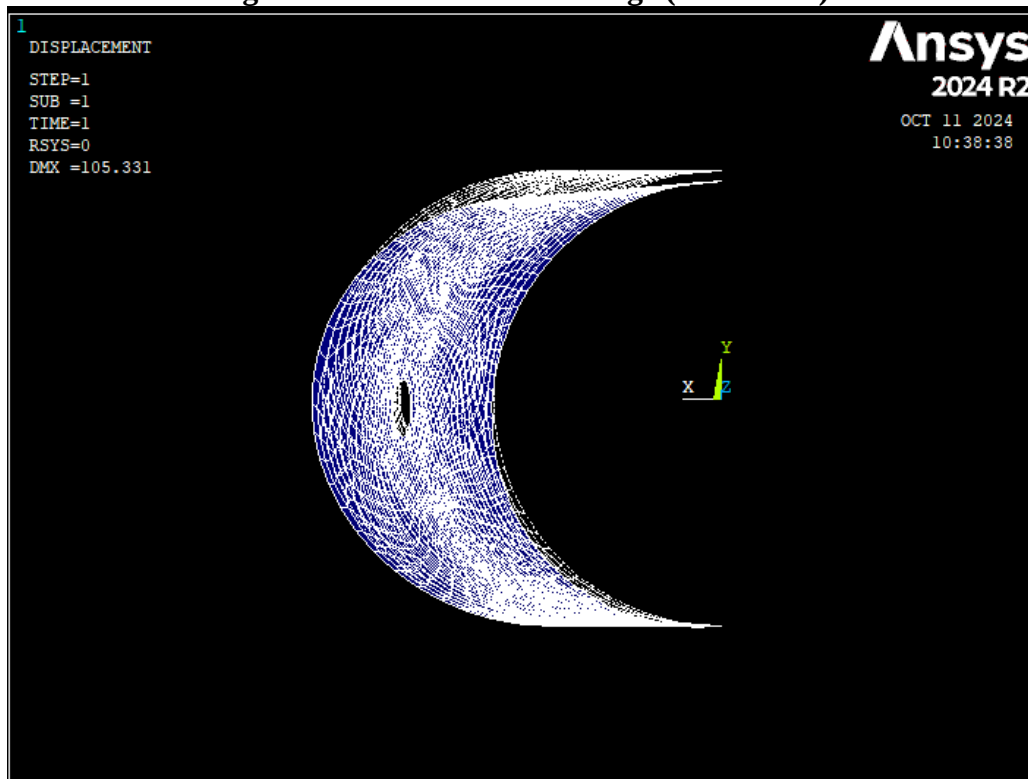


Figure 3: Element Coordinate System

**Figure 4: Element Coordinate Vector****Figure 5: Stresses of the Fuselage (Von Mises)**

**Figure 6: Strains of the Fuselage (Von Mises)****Figure 7: Displacements of the Fuselage**

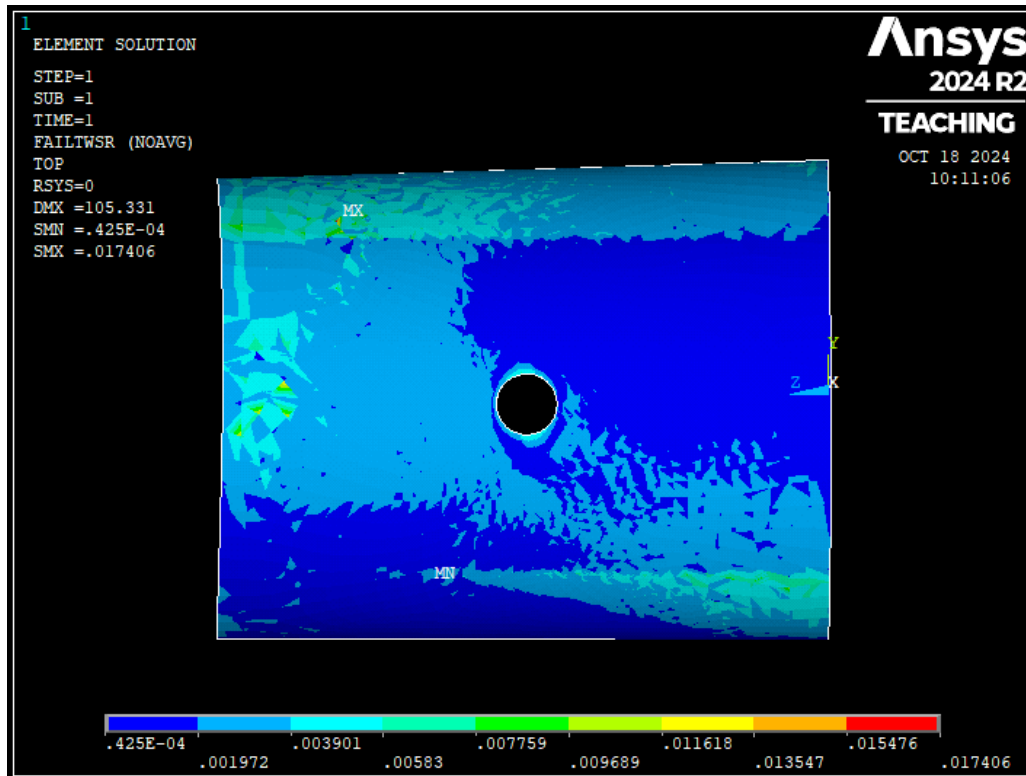


Figure 8: Overall Failure Contour Index

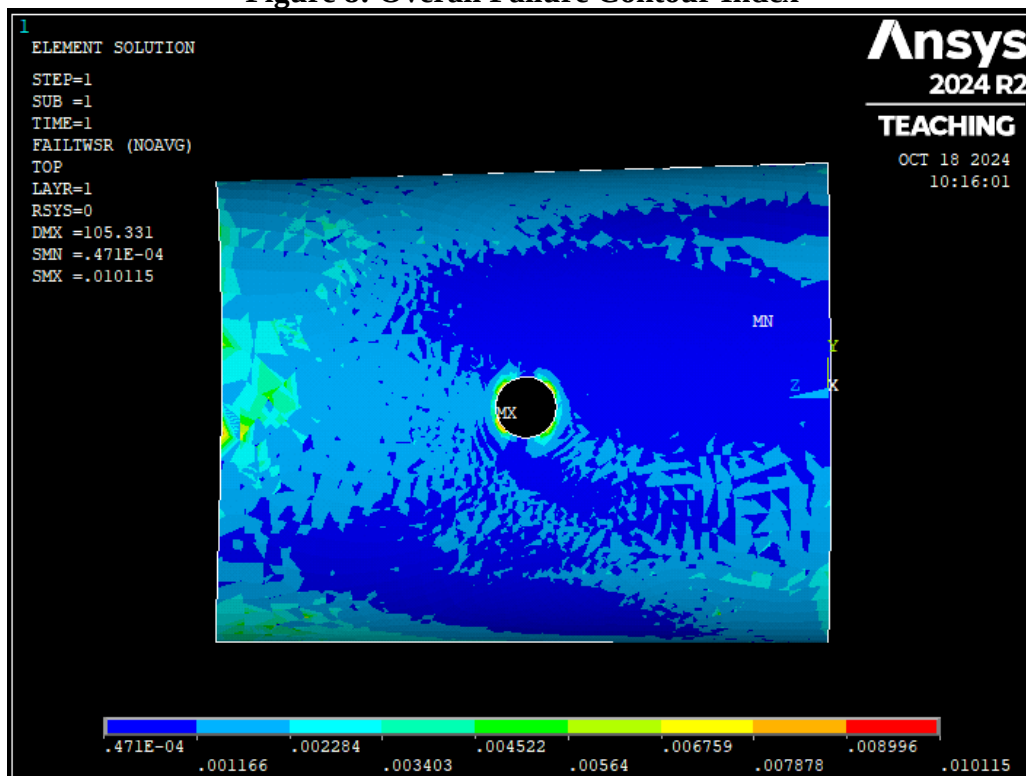


Figure 9: Individual Layer Failure Criteria (0°)

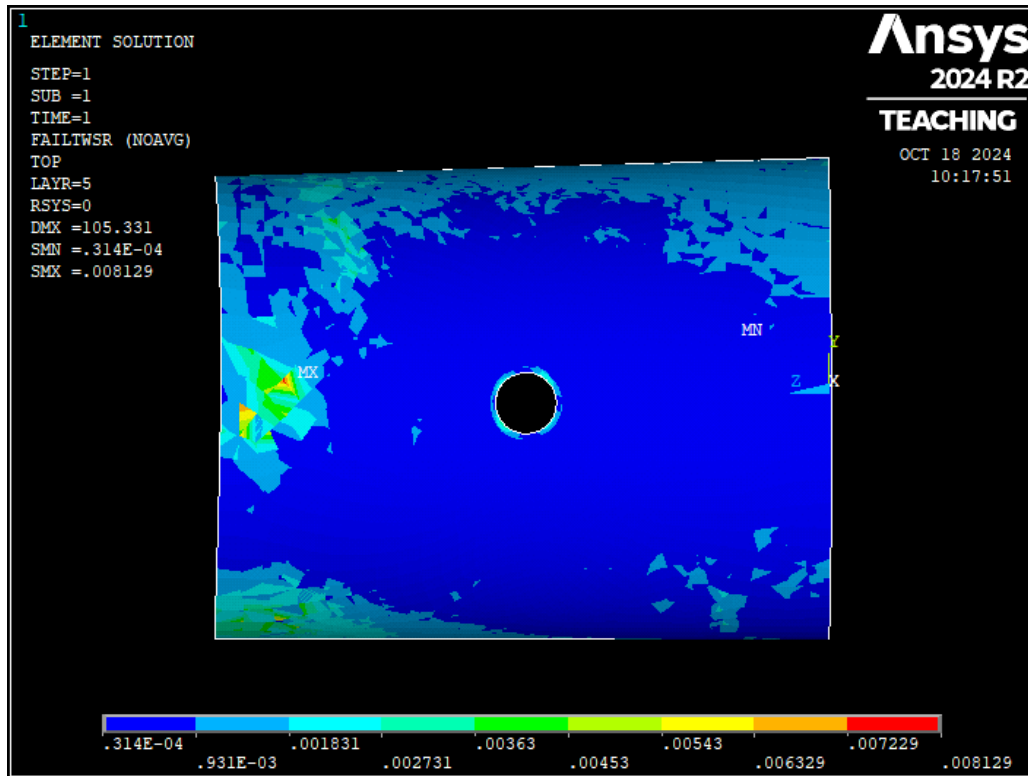


Figure 10: Individual Layer Failure Criteria (45°)

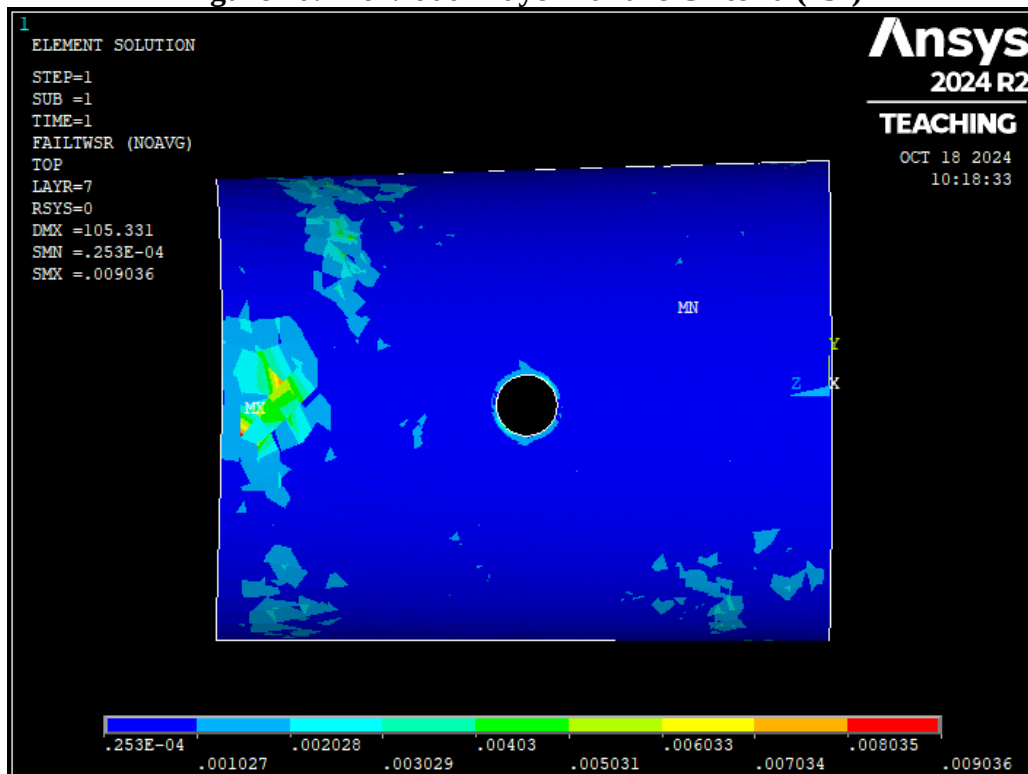


Figure 11: Individual Layer Failure Criteria (-45°)



## **PART 2**

### **Introduction**

For part 2, we are prompted to create and analyze a plate with an elliptical hole subjected to an axial distributed load of 10 kPa. The plate material is a layered composite with  $t = 125\text{e-}6\text{m}$  and ply sequence  $[0_4/45_4]$ . The elastic constant and failure characteristics are the same as the defined in Part 1.

### **Objectives**

- Convergence plot showing the maximum von Mises stress with respect to number of elements.
- Displacement plot.
- Output contour plots of:
  - Von Mises stress in layers 3 and 6.
  - Inverse Tsai-Wu for layers 3 and 6.
- Table with maximum von Mises stress in layers 3 and 6.



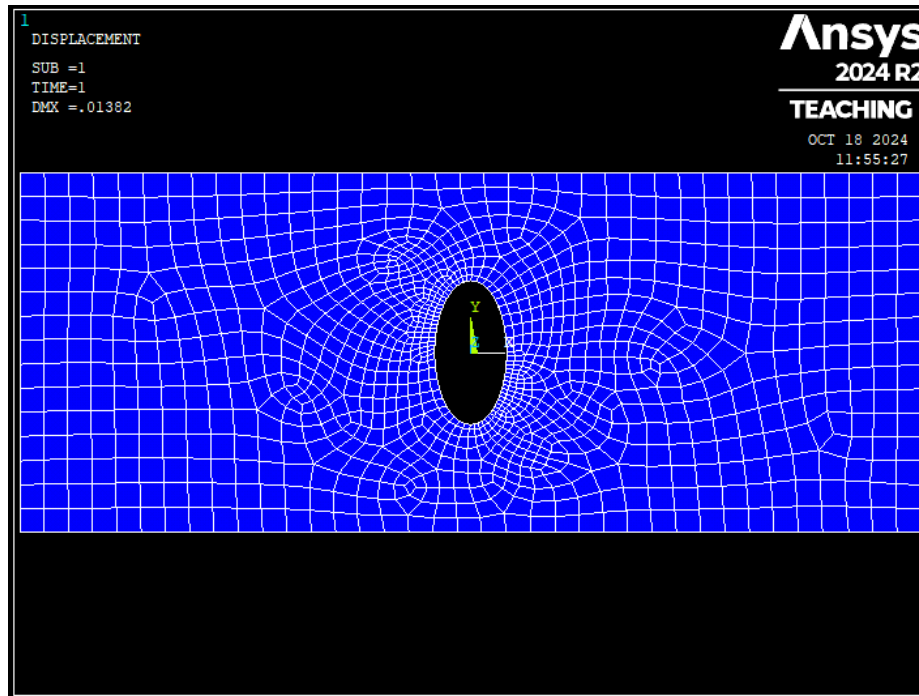


Figure 3: Displacements of the Plate (Front View)

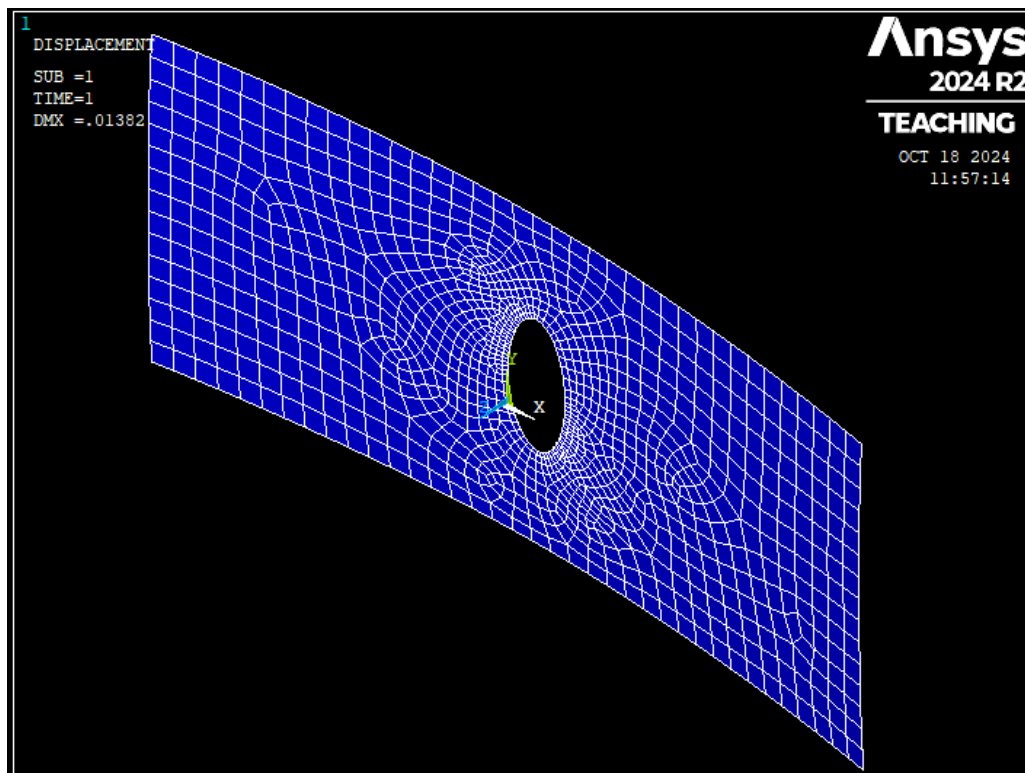


Figure 4: Displacements of the Plate (Perspective View)

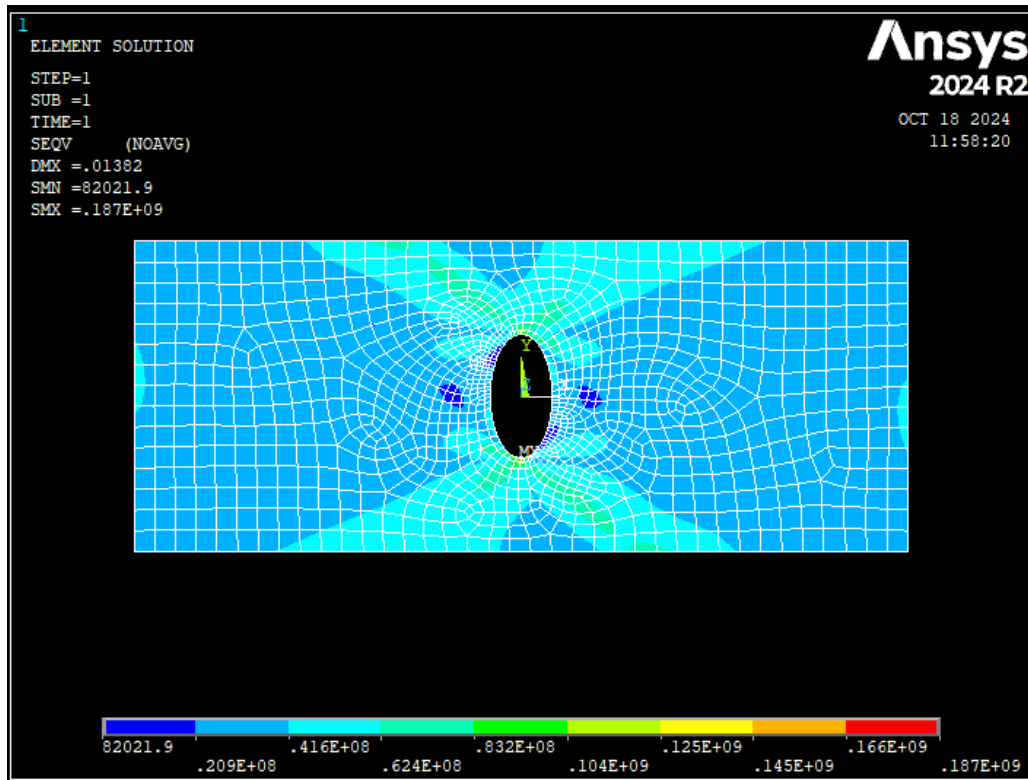


Figure 5: Stresses of the Plate (Von Mises)

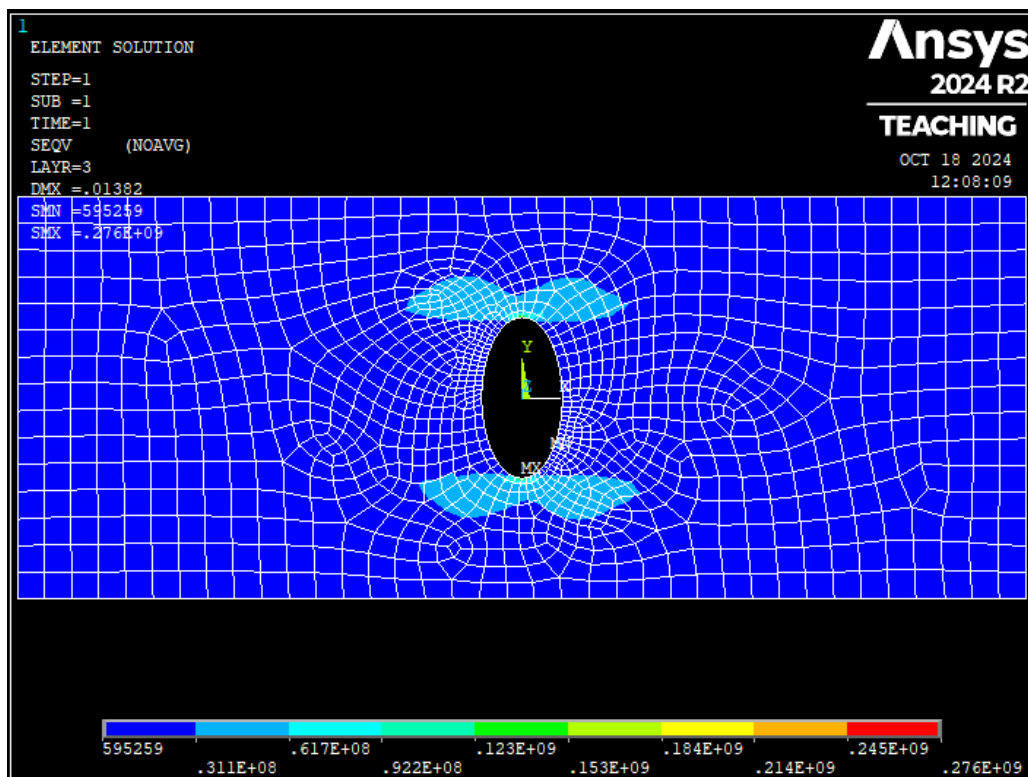


Figure 6: Von Mises Stress in Layer 3 (0°)

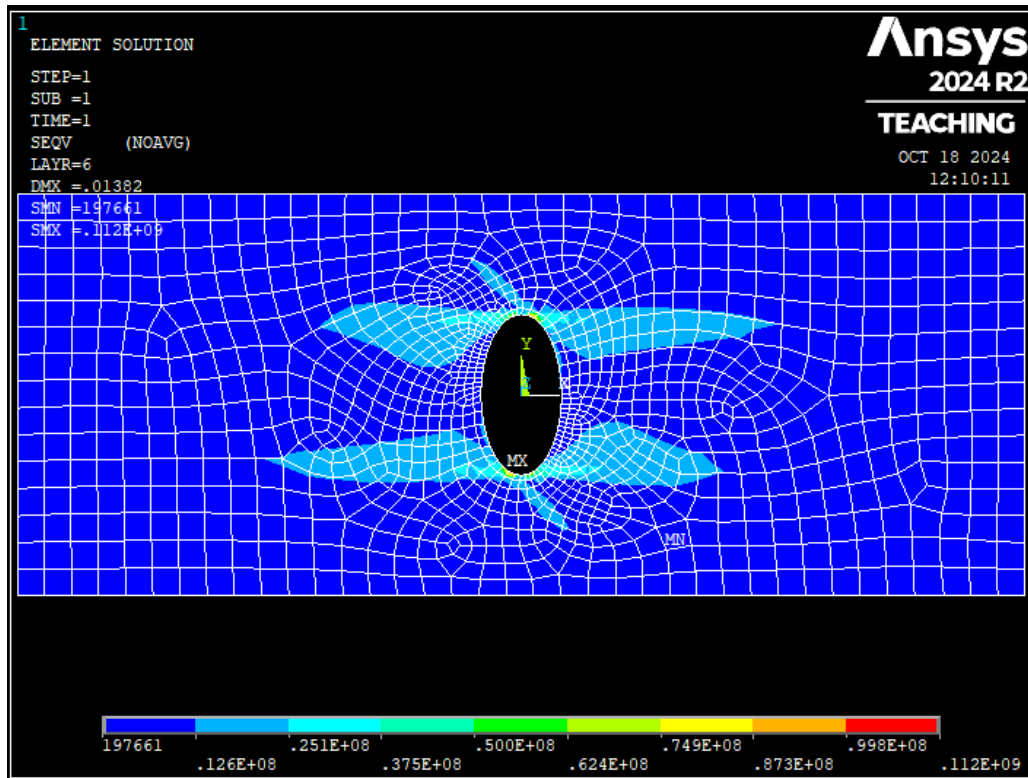


Figure 7: Von Mises Stress in Layer 6 (45°)

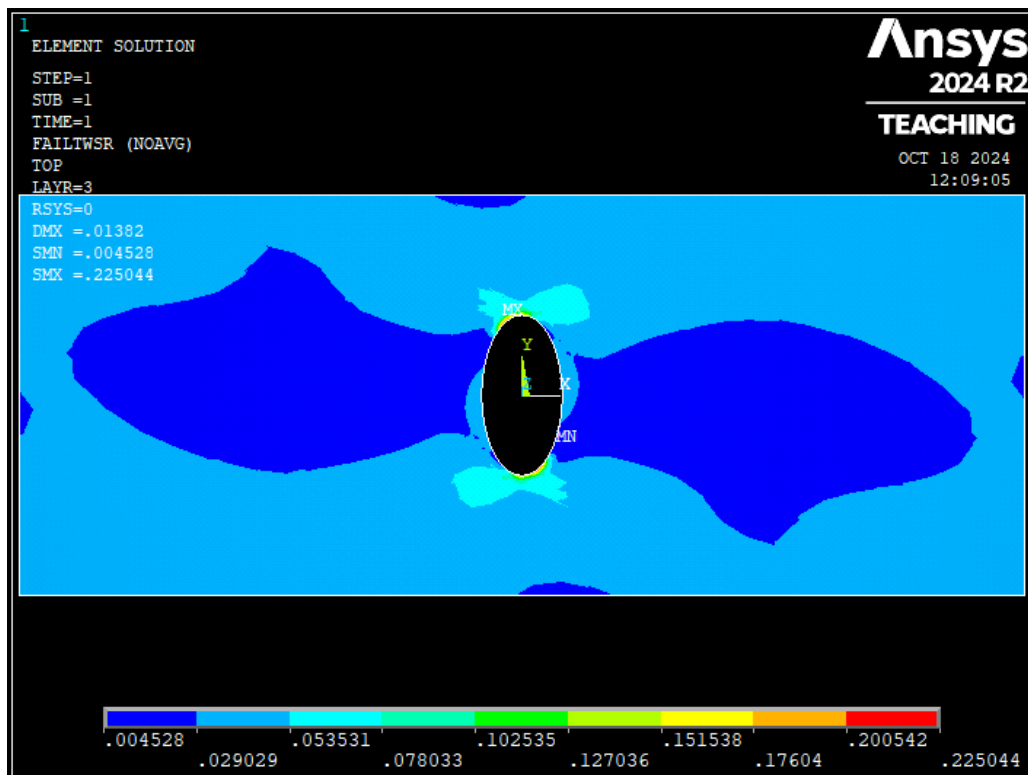


Figure 8: Inverse Tsai-Wu Index for Layer 3 (0°)

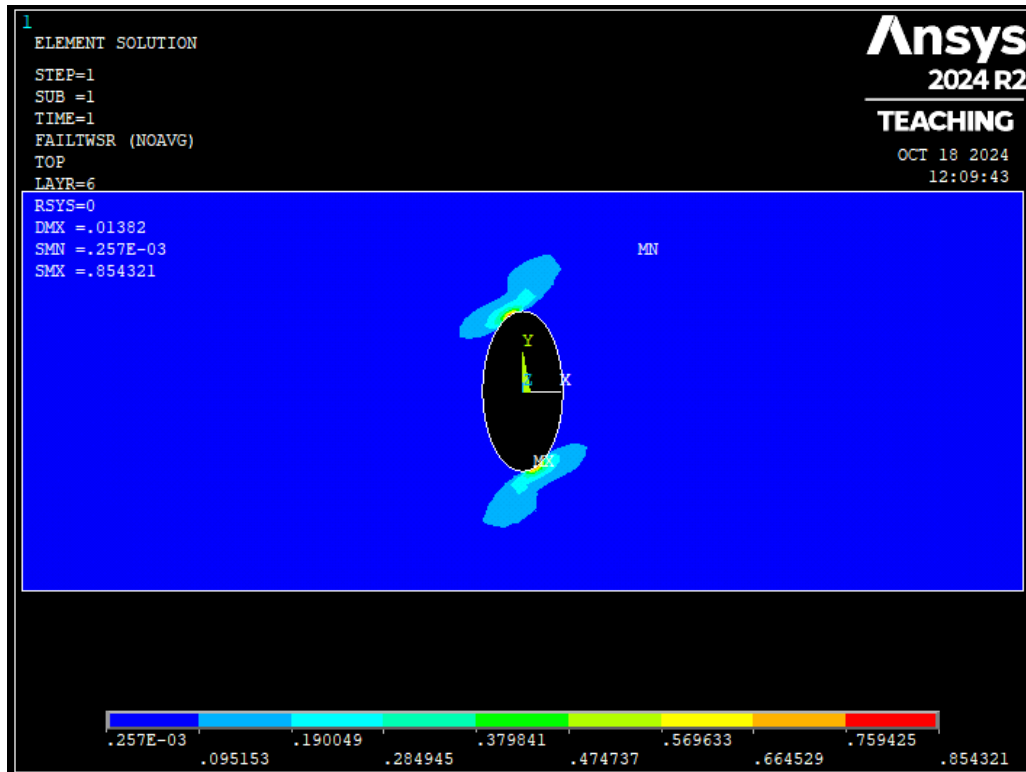


Figure 9: Inverse Tsai-Wu Index for Layer 6 (45°)

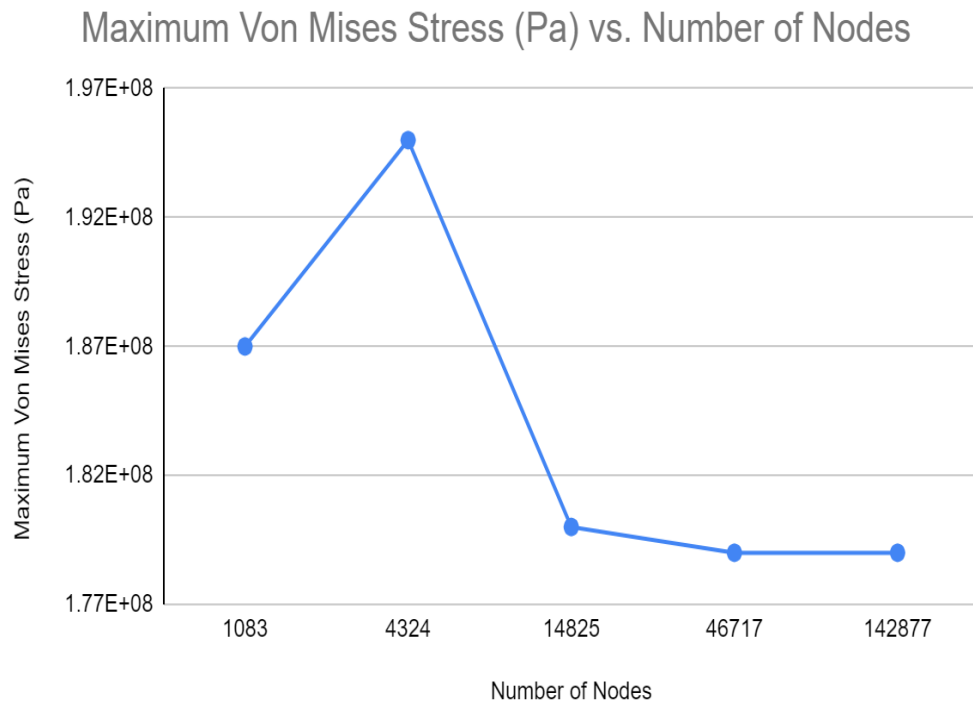


Figure 10: Convergence plot showing the maximum von Mises stress with respect to number of elements

| Layer | Max Von Mises Stress (Pa) |
|-------|---------------------------|
| 3     | .276E9                    |
| 6     | .112E9                    |

**Table 1: Maximum von Mises stress in layers 3 and 6**

## Analysis

### Will the plate fail?

- The plate has .854 failure criteria on layer 6, compared to layer 3, which as a failure criterion of .225. This means that the most likely area to fail is within layer 6 near the top left and bottom right of the circle. Even though this is the most likely area to fail, it's not 1, meaning that at the moment it will not fail and will need an increase of tension to do so.

Appendix:

/clear

/PREP7

ET,1,SHELL281

MPTMP,,,,,,,,

MPTMP,1,0

MPDATA,EX,1,,154E9

MPDATA,EY,1,,14.79E9

MPDATA,EZ,1,,14.79E9

MPDATA,PRXY,1,,0.175

MPDATA,PRYZ,1,,0.3

MPDATA,PRXZ,1,,0.175

MPDATA,GXY,1,,7.48

MPDATA,GYZ,1,,3.37

MPDATA,GXZ,1,,7.48

FC,1,S,XTEN,1500E6

FC,1,S,YTEN,40E6

FC,1,S,ZTEN,40E6

FC,1,S,XCMP,-1500E6

FC,1,S,YCMP,-246E6

FC,1,S,ZCMP,-246E6

FC,1,S,XY,68E6

FC,1,S,YZ,68E6

FC,1,S,XZ,68E6

FC,1,S,XYCP,

FC,1,S,YZCP,

FC,1,S,XZCP,

FC,1,EPEL,XTEN,

FC,1,EPEL,YTEN,

FC,1,EPEL,ZTEN,

---

```
FC,1,EPEL,XCMP,  
FC,1,EPEL,YCMP,  
FC,1,EPEL,ZCMP,  
FC,1,EPEL,XY,  
FC,1,EPEL,YZ,  
FC,1,EPEL,XZ,  
sect,1,shell,,  
secdata, 125E-6,1,0.0,3  
secdata, 125E-6,1,0.0,3  
secdata, 125E-6,1,0.0,3  
secdata, 125E-6,1,0.0,3  
secdata, 125E-6,1,45,3  
secdata, 125E-6,1,45,3  
secdata, 125E-6,1,45,3  
secdata, 125E-6,1,45,3  
secoffset,MID  
seccontrol,,,, , , ,
```

```
KEYOPT,1,1,0  
KEYOPT,1,5,0  
KEYOPT,1,8,1  
KEYOPT,1,9,0  
KEYOPT,1,10,0  
KEYOPT,1,11,0
```

```
k,1,-0.125,-0.05,0  
k,2,-0.125,0.05,0  
k,3,0.125,0.05,0  
k,4,0.125,-0.05,0
```

```
k,5,0.02,0  
k,6,0,0.02,0  
k,7,-0.02,0  
k,8,0,-0.02,0  
k,9,0,0
```

```
L,1,2  
L,2,3  
L,3,4  
L,4,1
```

```
LARC,5,6,9,0.02  
LARC,6,7,9,0.02  
LARC,7,8,9,0.02  
LARC,8,5,9,0.02  
FLST,2,4,4,ORDE,2  
FITEM,2,5  
FITEM,2,-8  
LSSCALE,P51X, , ,0.5,1,1, ,1,1  
AL,ALL
```

```
DK,1,UX  
DK,1,UY  
DK,1,UZ
```



```
DK,2,UX
DK,2,UZ
DK,3,UZ
```

```
SFL, 3, PRES, -10000
SFL, 1, PRES, -10000
```

```
AMESH,ALL
```

```
ACLEAR,      1
CSWPLA,11,0,1,1,
FLST,2,1,5,ORDE,1
FITEM,2,1
CM,_Y,AREA
ASEL, , , ,      1
CM,_Y1,AREA
CMSEL,S,_Y
!*
CMSEL,S,_Y1
AATT,      1, ,      1,      11,      1
CMSEL,S,_Y
CMDELE,_Y
CMDELE,_Y1
!*
SMRT,6
SMRT,5
SMRT,4
SMRT,3
/UI,MESH,OFF
MSHAPE,0,2D
MSHKEY,0
!*
CM,_Y,AREA
ASEL, , , ,      1
CM,_Y1,AREA
CHKMSH,'AREA'
CMSEL,S,_Y
!*
AMESH,_Y1
!*
CMDELE,_Y
CMDELE,_Y1
CMDELE,_Y2
!*
! /VIEW,1,1,1,1
! /ANG,1
! /REP,FAST
! /VIEW,1,,1
! /ANG,1
! /REP,FAST
FINISH

/SOL
antype, 0
SOLVE
FINISH
```

```
/POST1  
PLESOL,S,EQV,0,1.0
```

```
/PREP7  
LREFINE,ALL,,,2
```

```
/SOL  
antype, 0  
SOLVE  
FINISH  
/POST1  
PLESOL,S,EQV,0,1.0
```